Combining the Convolution and Transformer for Classification of Smoke-Like Scenes in Remote Sensing Images

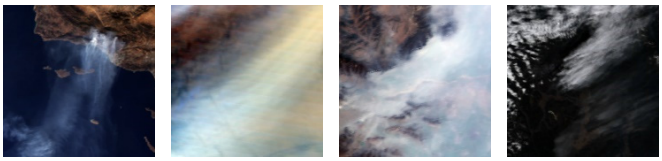
Shikun Chen, Weixuan Li, Yichao Cao, and Xiaobo Lu

***Abstract*— Remote sensing (RS) images are used in a wide range of tasks. In the fire detection field, smoke in RS images is considered as an indicator of wildfires. However, smoke-like scenes, e.g., cloud, in RS images increase the difficulty of smoke recognition. Convolutional neural networks (CNNs) have greatly promoted the development of image processing. CNNs are good at capturing local features; however, their ability to capture global features is relatively weak. Recently, the transformer deep learning model has shown strong potential in vision tasks. The transformer model utilizes self-attention modules to extract global features but may lose local details. Recognition of smoke in RS images depends strongly on the combination of both local and global features. Thus, this article proposes the transformer enhanced convolutional network (TECN) to classify RS smoke-like scenes. The proposed hybrid TECN model exploits the advantages of the CNN and transformer techniques at the same time. In TECN, the feature merge and intelligent aggregation modules are used to promote conversion and aggregation between CNN feature maps and transformer patch embeddings. Experiments are conducted on the USTC_SmokeRS dataset, which is developed for the classification of RS smoke-like scenes. The experimental results demonstrate that the proposed TECN achieves a competitive accuracy of 98.39% on this dataset.**

***Index Terms*— Convolutional neural network (CNN), scene classification, smoke-like scenes, transformer.**

I. Introduction

THE spatial and temporal resolutions of remote sensing (RS) images are improving constantly, and RS images are widely used in various tasks. One important task is the monitoring and early identification of fires. Fire disasters pose serious threat to the natural environment and human life; thus, early detection of fire is of great importance. Previously, people used surveillance video to monitor fire disasters. However, the scope of surveillance video is relatively small, and its costs are high, which often lead to missing the best time to perform fire monitoring. In contrast, RS images have wide coverage, low cost, and good timeliness. When a fire breaks out, satellite sensors identify the smoke emitted by the fire, and smoke in RS images is considered as an indicator of fire disasters.



(a) (b) (c) (d)

Fig. 1. Smoke-like scenes in RS images: (a) smoke; (b) dust; (c) haze; and (d) cloud.

Real-time monitoring and identification of fire disasters can be realized by recognizing smoke in RS images. However, this is a difficult task because several smoke-like scenes can be found in RS images. For example, clouds, which are very similar to smoke in terms of color, shape, and texture, are common in RS images. Other smoke-like scenes, e.g., dust and haze, also appear frequently in RS image (Fig. 1). It is easy to visually confuse such smoke-like scenes with real smoke, which increases the difficulty of smoke recognition and fire detection [1].

However, computer vision technologies help us understand images. For example, scene classification is a basic task of such technologies. In scene classification, the goal is to assign a semantic label to each image, and scene classification can distinguish smoke from other smoke-like scenes. Once an RS image is labeled as a smoke scene, there may be a fire disaster in corresponding area. Previously, traditional methods used low-level features as prior knowledge to investigate images; however, it is difficult for such methods to capture deep features. Due to the developments in computational power and data annotation techniques, convolutional neural networks (CNNs) have demonstrated success in computer vision tasks. For example, the AlexNet [2], VggNet [3], ResNet [4], DenseNet [5], and other CNN models have improved classification accuracy on various datasets. In a CNN, convolution operations are employed to extract local features as feature representations in a hierarchical manner [6]. CNNs are effective at capturing local features; however, their ability to extract global features is relatively weak. Some attempts have been made to break through this limitation, e.g., increasing the network depth and enlarging the receptive field; however, to date, the results have not been sufficient. As shown in Fig. 2, sometimes smoke occupies a very small

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Shikun Chen, Yichao Cao, and Xiaobo Lu are with the School of Automation, Southeast University, Nanjing 210096, P. R. China, and also with the Key Laboratory of Measurement and Control of Complex Systems of Engineering, Ministry of Education, Nanjing 210096, P. R. China (e-mail: 230198550@seu.edu.cn; yichaocao@seu.edu.cn; xblu2013@126.com).

Weixuan Li is with Nanjing Enbo Technology Company Ltd., Nanjing 210014, China (e-mail: 676354385@qq.com).

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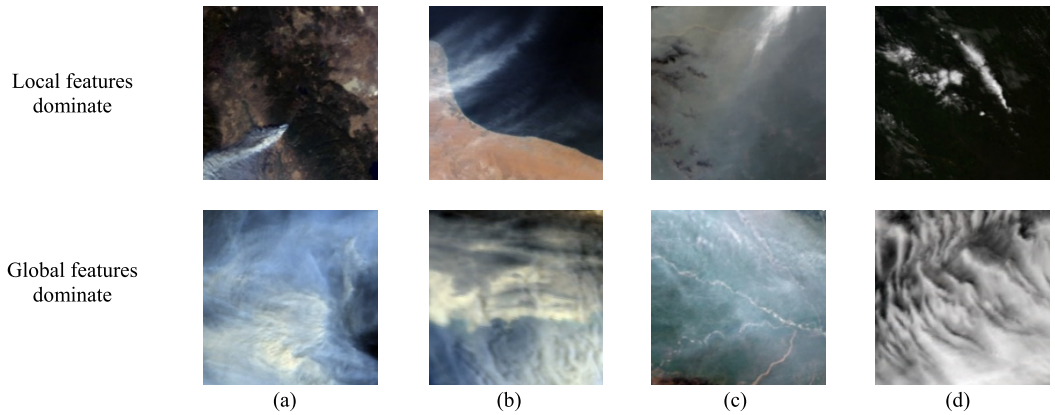


Fig. 2. Large intraclass variance of RS images: (a) smoke; (b) dust; (c) haze; and (d) cloud.

region in RS images. In such cases, local features play a leading role in understanding images. CNNs can perform well when processing such images. However, smoke may fill the entire image in other case, and global features are required to analyze such images. Thus, both local and global features are essential for recognizing smoke and distinguishing smoke from smoke-like scenes, and improving a CNN's ability to capture global features is a key to improving its performance. The transformer model [7] was originally proposed for natural language processing (NLP) tasks. Rather than convolution operations, transformer uses self-attention to extract features, where the self-attention mechanism can obtain the long-range dependencies among different words in a sentence and learn the associative representations across sequences. Inspired by NLP transformers, Dosovitskiy *et al.* [8] proposed the vision transformer (ViT) to handle vision tasks. In ViT, an input image is split into nonoverlapping image patch embeddings (similar to words in a sentence), and cascaded multihead self-attention (MSA) blocks capture parameterized vectors as feature representations. As a result, ViT can capture long-range feature dependencies among patch embeddings, and these dependencies constitute global features. Thus, the CNN and transformer techniques are complementary, i.e., a CNN focuses on local details but has difficulty exploring global features, and transformer has advantages in terms of global feature extraction but lacks local details. By combining the CNN and transformer techniques, both local and global features can be retained simultaneously, which could improve smoke recognition in RS images.

Thus, in this article, we propose a transformer enhanced convolutional network (TECN) for the classification of RS smoke-like scenes. As shown in Fig. 3, the proposed TECN involves two main components, i.e., a CNN feature extractor (blue dashed line) and a transformer pathway (green dashed line). The four-stage CNN feature extractor captures local details from CNN tensors. The transformer pathway comprises four transformer stages and is designed to provide global information to the CNN feature extractor. When an image is input to the proposed TECN, convolution with stride 2 and pooling operations are performed to initially process the image. The processed image is then input to the feature extractor and transformer pathway separately. Note that the dimensions of the CNN's feature maps and transformer patch embeddings differ; thus, there is a distinct semantic gap between them. A direct and rough conversion between feature maps and patch embeddings could lead to feature loss and vague classification. To allow the conversion process between feature maps and patches to retain the advantages of the CNN and transformer techniques, we designed the feature merge module (FMM) and intelligent aggregation module (IAM). The FMM and IAM take inputs from both the CNN feature extractor and transformer pathway. In the FMM, feature maps from the CNN stage are converted to transformer space and then merged with patches from the transformer stages. Rather than performing a direct conversion, global pooling is first employed as a gate to select the informative channels of the input feature maps. Here, channels containing more class-specific features are highlighted. As a result, the transformation from feature maps to patches is much smoother. The guidance of local details provided by the CNN can be propagated to the transformer patches. Then, the class token of the transformer input and patches from the CNN input are fused as the output of the FMM. In the IAM, after the transformer input is reshaped to feature maps, it then goes through an average pooling. Here, average pooling computes the average value in a pooling kernel and promotes the global features obtained in the transformer stages. We then employ a channelwise attention operation to highlight the class-sensitive channels of the pooled features [9]. Then, the transformer input is aggregated with the feature maps from the CNN adaptively. Long-range dependencies between visual elements from transformer are used to enhance the CNN in the aggregation. The aggregated features contain local details and retain global features. Due to this hybrid structure, the proposed TECN can perform RS smoke-like scene classification effectively.

Our primary contributions are summarized as follows.

1) The TECN, which extracts local features and retains global features simultaneously, is proposed for the classification of smoke-like scenes in RS images.
2) The FMM is proposed to promote the conversion from CNN feature maps to patch embeddings.
3) The IAM is proposed to make the aggregation between feature maps and patch embeddings an adaptive and intelligent process.

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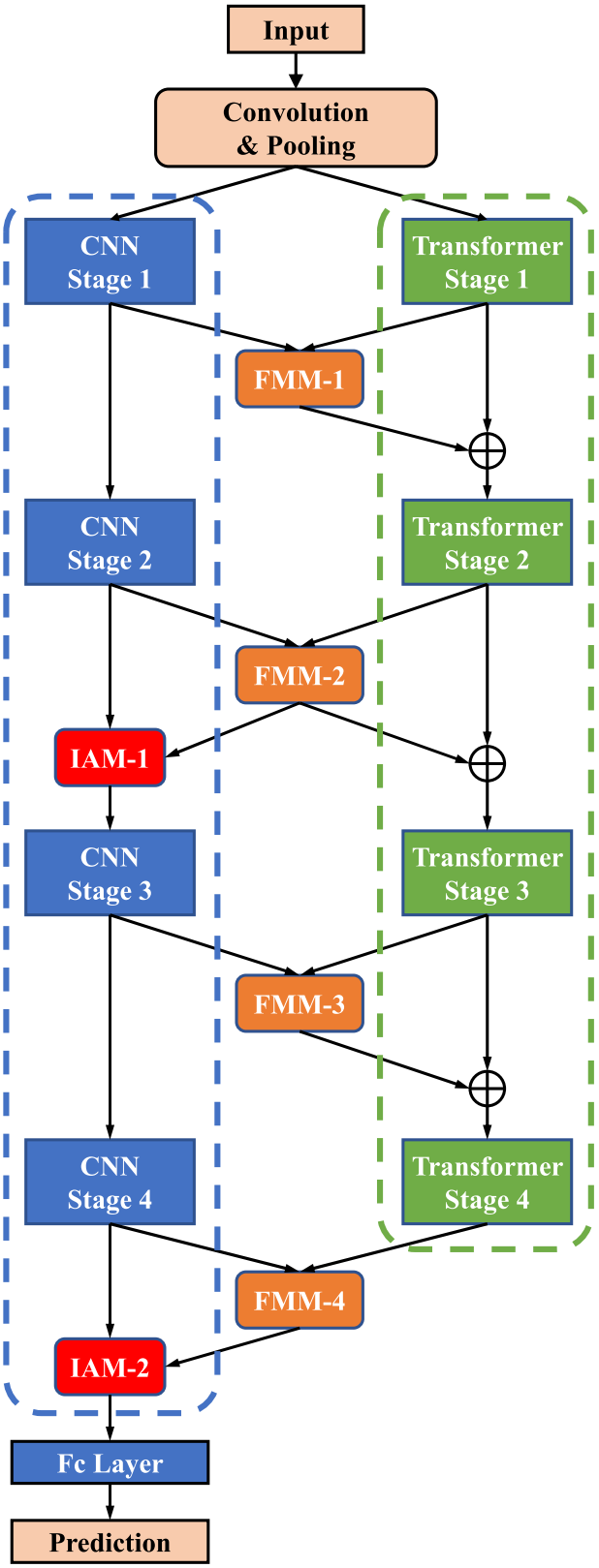


Fig. 3. Structure of the proposed TECN.

4) The proposed TECN achieves a competitive classification accuracy of 98.39% on the USTC_SmokeRS dataset.

II. Related Work

A. Smoke Recognition

Algorithms based on low-level features have been used to recognize smoke. Visual interpretation methods select three spectral bands of satellite sensors to generate true-color or false-color images to manually determine the presence of smoke in an image [10], [11]. However, such methods cannot realize automatic recognition. Multithreshold methods detect smoke by extracting optimal thresholds for the reflectance or brightness of spectral bands [12], [13]. Recently, deep learning techniques have boosted smoke recognition with the development of artificial intelligence. Yin *et al.* [14] designed the deep normalization and CNN (DNCNN) to alleviate the gradient dispersion and overfitting problems in video-based smoke detection. In addition, the deep convolutional generative adversarial networks (DCGAN) [15] was designed to realize real-time smoke recognition using motion-based geometrical image transformation. Ba *et al.* [1] employed scene classification technology to distinguish smoke from other scenes. They also established the USTC_SmokeRS dataset of smoke-like scenes and proposed a baseline model for their dataset, i.e., SmokeNet.

B. RS Scene Classification

The scene classification task attempts to interpret an image by applying a semantic label. Handcrafted feature models have been applied in RS scene classification. For example, scale-invariant feature transform [16] and bag-of-visual-words (BoVW) [17], [18] are widely used methods. Zou *et al.* [19] combined multiple types of handcrafted features, and Cheng *et al.* [20] embedded convolutional features into BoVW to classify RS images. However, handcrafted feature-based methods exhibit weak feature representation.

Due to rapid developments in terms of computational power, deep learning models are being used more frequently to handle RS scene classification. Cheng *et al.* [21] provided a thorough survey of deep learning techniques for RS scene classification. In addition, the main challenges facing RS scene classification, mainstream RS scene classification benchmarks, and related deep learning-based approaches have been reviewed systematically. Adaptive balanced network (ABNet) [22] was designed to solve the problem of complicated background and imbalanced object scales in RS images, and has shown outperforming performances. Wang *et al.* [23] proposed a hybrid model named HFANet which can effectively detect salient objects with complicated edges and irregular topology in RS images. Nogueira *et al.* [24] explored three training strategies for RS scene classification. To correct potentially incorrect labels, Li *et al.* [25] proposed an RS image scene classification-oriented (RSSC-oriented) error-tolerant deep learning approach called RSSC-ETDL. In addition, Chen *et al.* [26] employed knowledge distillation to improve classification performance. The discriminative CNN (DCNN) model [27], which was trained by optimizing a discriminative objective function, was proposed for RS scene classification. Miao *et al.* [28] proposed a novel graph-based semisupervised learning model to deal with hyperspectral

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imagery classification. Key filter bank network (KFBNet) [29] employed a filter bank to aggregate features in RS images. Xu *et al.* [30] proposed a dual-branch structure to explore crucial areas. Chen *et al.* [31] presented the self-adaptive feature aggregation (SAFA) model, which aggregates different features for RS scene classification. In addition, SmokeNet [1] was specifically designed for smoke-like scene classification in RS images.

C. Vision Transformer

Vaswani *et al.* [7] proposed the self-attention mechanism to tackle NLP. Inspired by Vaswani *et al.* [7], Dosovitskiy *et al.* [8] introduced transformer to vision tasks and demonstrated that the ViT could achieve remarkable performances on large datasets compared to CNN models. In addition, the Swin transformer [32] employed shifted windows and a hierarchical structure to reduce computational costs. Data-efficient image transformers (DeiT) [33] employed a novel distillation procedure and training strategy such that it did not require large amounts of training data. Tokens-to-token-ViT (T2T-ViT) [34] progressively tokenized images and could be trained from scratch. However, the self-attention mechanism in transformer models may lead to the loss of spatial local details. Some studies have attempted to combine local features from a CNN and global features from a transformer model. For example, the MobileViT model [35] employed a lightweight structure and learns global representations using transformers as convolutions. In addition, conformer [6] employed a structure that combines CNN and transformer models. DEtection TRansformer (DETR) [36] sent feature maps to a transformer encoder–decoder architecture for objection detection.

III. Methodology

For smoke-like scene classification in RS images, exploring only local or global features may work well in some cases. In this article, we propose TECN to further improve classification performance. TECN is a hybrid model that exploits the benefits of the CNN and transformer. The structure of the proposed TECN is shown in Fig. 3. TECN comprises two main components, i.e., a CNN feature extractor (blue dashed line) and a transformer pathway (green dashed line), both of which have four stages. Like most neural networks, a convolution operation and max pooling are initially used to extract features from the input. Max pooling can retain the most important features in the pooling kernel and highlight local features. After pooling, features maps are input to the CNN feature extractor and transformer pathway separately. In the CNN feature extractor, convolution operations are cascaded to collect spatial details. For the transformer pathway, self-attention modules are stacked to capture global features in the patch embeddings. Here, functions of CNN and transformer are complementary with each other. After each stage of the CNN, feature maps rich in local features are sent to the transformer pathway to provide the spatial inductive bias ignored by the transformer. Here, the FMM promotes smooth conversion from feature maps to patch embeddings and combines features from two sources. The feature maps are converted to patch embeddings in the FMM and sent to the next transformer stage. Then, transformer's self-attention models long-range, nonlocal dependencies. After leaving the transformer stages, the patch embeddings are fed back to the CNN feature extractor through the FMM and fused with feature maps in the IAM. The IAM aggregates local features from the CNN and global features from the transformer intelligently, which means IAM can select the most useful information for aggregation. Then, global features from the transformer help enhance the feature representations of the CNN feature extractor. The output of IAM-2 is sent to a classifier to generate the proposed TECN's final prediction.

TABLE I
Structure of DenseNet-121

Stages	Output size	DenseNet-121
Convolution	128 × 128	7×7 conv, stride 2
Pooling	64 × 64	Max pool, stride 2
Stage-1	32 × 32	$\begin{bmatrix} 1\times1 \text{ conv} \\ 3\times3 \text{ conv} \end{bmatrix} \times 6$
Stage-2	16 × 16	$\begin{bmatrix} 1\times1 \text{ conv} \\ 3\times3 \text{ conv} \end{bmatrix} \times 12$
Stage-3	8 × 8	$\begin{bmatrix} 1\times1 \text{ conv} \\ 3\times3 \text{ conv} \end{bmatrix} \times 64$
Stage-4	8 × 8	$\begin{bmatrix} 1\times1 \text{ conv} \\ 3\times3 \text{ conv} \end{bmatrix} \times 48$
Pooling	1 × 1	Global average pooling
Fc	1 × 1	1024 × num_class

A. CNN Feature Extractor

Some classical CNN models, e.g., ResNet and DenseNet, have demonstrated excellent representation capacity. During the training process, CNNs use convolution operations to capture local features in a hierarchical manner and retain local cues as feature maps [6]. However, training a model from scratch is relatively difficult, especially when only a small amount of data is available. Pretrained parameters can accelerate the training process and improve the performance [37]; thus, we adopt the pretrained policy. The weight parameters of the CNN feature extractor are initialized using the pretrained weight parameters of DenseNet-121 on ImageNet [38]. Note that we also considered employing other CNN models as our feature extractor, e.g., Pretrained ResNet, and the performance with different pretrained models is discussed in Section V. ResNet uses identity mapping to handle the gradient vanishing problem. Based on ResNet, DenseNet has fewer parameters and exhibits better performance. The input of each layer in DenseNet comprises all outputs of the former layers, and this connectivity allows each layer directly access to the gradients from the loss function and facilitates feature propagation along the network. The DenseNet-121 structure is detailed in Table I. As can be seen, in each stage of DenseNet, there is

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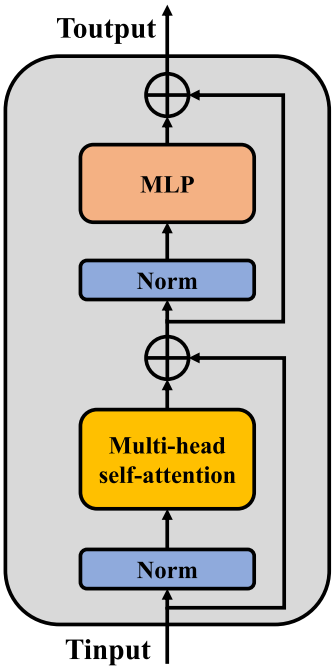


Fig. 4. Structure of the transformer block.

a stack of 1×1 and 3×3 convolution layers. The 1×1 convolution is introduced as a bottleneck layer prior to the 3×3 convolution to improve computational efficiency. Note that there is a transition layer after each 3×3 convolution. Here, each transition layer comprises a batch normalization (BN) layer and a 1×1 convolutional layer followed by a 2×2 average pooling layer [5].

B. Transformer Pathway

The transformer pathway is designed to provide further global guidance for progressive features from the CNN feature extractor. Prior to the first transformer stage, we perform convolution operations to project the tensor to a high-dimensional space with 768 channels. As a result, the transformer can use local information from the convolutions. This also aligns with the observation that introducing locality in early layers is beneficial to feature representation in transformers [39], [40]. In each transformer stage, the feature maps from the CNN of size $B \times C \times H \times W$ are flatten to patches of size $B \times (H \times W) \times C$, where B is the batch size, C is the number of channels, and H and W are the height and width of the feature maps, respectively. This means that each pixel is taken as a single patch embedding. The patch sequence is obtained by flattening the spatial dimensions of the feature map and then projected to the transformer dimension. We also attach a trainable class token to these patches. This class token obtains class-specific information during training and is used to make the final prediction in the standard ViT [8]. In the proposed method, this trainable token is combined with feature maps in the FMM to provide global features.

There are four transformer stages in the TECN, and each stage contains two repeated transformer blocks. The structure of the transformer block is shown in Fig. 4. As can be seen, each transformer block comprises MSA layers and multilayer perceptron (MLP) blocks. Here, layer norm is applied before each MSA and MLP. In MSA, the input of size $B \times (H \times W) \times C$ is transformed to a query: $Q \in B \times (H \times W) \times D_{\mathrm{q}}$, key $K \in B \times (H \times W) \times D_{\mathrm{q}}$, and value $V \in B \times (H \times W) \times D_{\mathrm{V}}$ matrices. Here, C, D_{q}, and D_{V} are the dimensions of the input, query matrix, and value matrix, respectively. Then, the self-attention is formulated as follows:

$$\text{Attention}\ (Q, K, V) = \sigma\left(\frac{QKT}{\sqrt{D_{\mathrm{V}}}}\right)V \tag{1}$$

where $\sigma(\cdot)$ denotes the SoftMax function. Self-attention is performed h times to form the MSA layer, where h is the number of heads (h is set to 12 in this article). The output of MSA is fed into the MLP block, which comprises two fully connected layers separated by a Gaussian error linear unit (GELU) activation function. In the MLP, patch embeddings are up-projected to 3072 dimensions by the first fully connected layer and then down-projected to 768 dimensions by the second fully connected layer. The entire process in a transformer block can be expressed as follows:

$$T_{\text{output}}^{\text{MSA}} = \text{MSA}\left(\tau\left(T_{\text{input}}\right)\right) + T_{\text{input}} \tag{2}$$

$$T_{\text{output}} = \text{MLP}\left(\tau\left(T_{\text{output}}^{\text{MSA}}\right)\right) + T_{\text{output}}^{\text{MSA}} \tag{3}$$

where T_{input} is the input to the transformer stage, $\tau(\cdot)$ is the layer norm function, $T_{\text{output}}^{\text{MSA}}$ represents the output of the MSA layer, and T_{output} denotes the output of the transformer blocks.

C. Feature Merge Module

The FMM bridges the semantic gap between the CNN and transformer. The sizes of CNN feature maps and transformer patch embeddings are inconsistent; thus, direct and rigid conversion between them may lead to the loss of useful information. The FMM can reduce misalignment between local features from the CNN feature extractor and global features from the transformer pathway. As shown in Fig. 5, the FMM takes two inputs, i.e., $F_{\text{input}}^{\text{C}}$ from the CNN feature extractor and $F_{\text{input}}^{\text{T}}$ from the transformer pathway. $F_{\text{input}}^{\text{C}}$ has a size of $B \times C \times H \times W$, and the size of $F_{\text{input}}^{\text{T}}$ is $B \times (1 + H \times W) \times C$, where B is the batch size, C is the number of channels, H and W are the height and width of the feature map, respectively, and 1 denotes the class token. Before being transformed to patch embeddings, a global pooling [1×1 convolution with BN and rectified linear unit (ReLU)] is performed on $F_{\text{input}}^{\text{C}}$, and its channel number is changed to 768. This global pooling is treated as a gate to select informative channels and make the input to be more class-sensitive. Then, average pooling with stride 2 is employed to downsample the pooled $F_{\text{input}}^{\text{C}}$. The average pooling computes the average value among adjacent pixels, which can help promote global information. As a result, the CNN input is more aligned with the transformer features. After average pooling, feature maps are reshaped to patch embeddings $F_{\text{patch}}^{\text{C}}$. $F_{\text{patch}}^{\text{C}}$ and transformer input $F_{\text{input}}^{\text{T}}$ are fused to obtain $F_{\text{output}}^{\text{T}}$. All four FMMs in the proposed TECN have $F_{\text{output}}^{\text{T}}$. As shown in Fig. 3, each $F_{\text{output}}^{\text{T}}$ of the FMM is added to the output from the former transformer stage, and their sum

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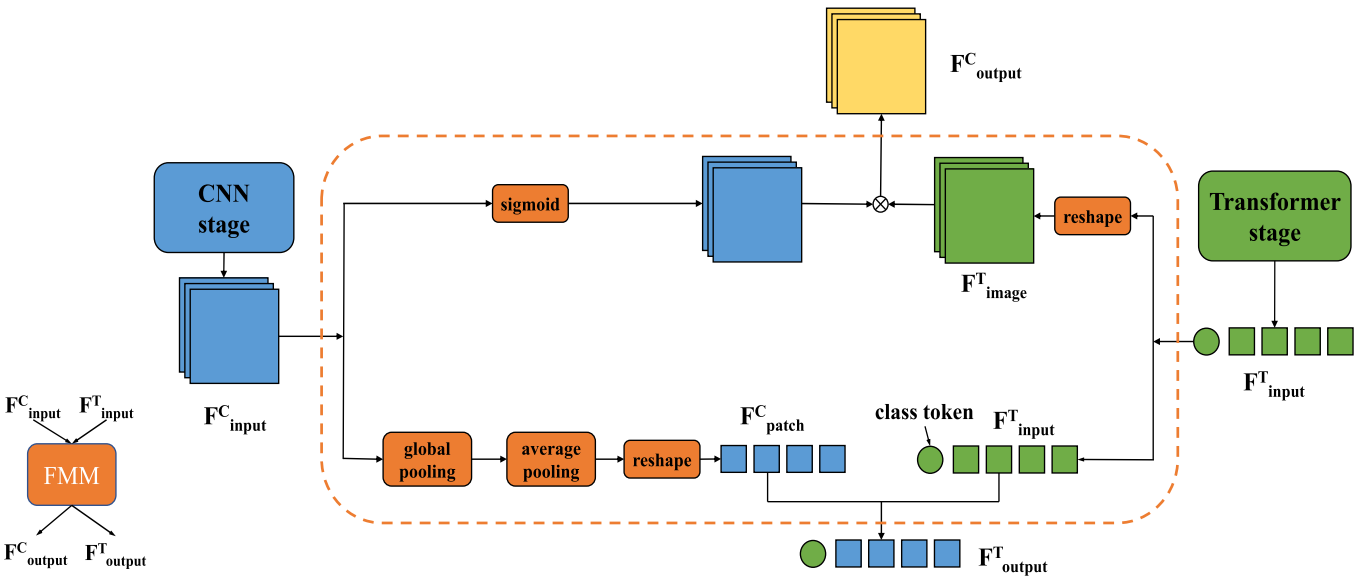


Fig. 5. Structure of FMM.

is input to the next stage. As shown in Fig. 5, during the fusion process, all patch embeddings of $F^{\mathrm{T}}_{\mathrm{input}}$ are abandoned except the class token. $F^{\mathrm{C}}_{\mathrm{patch}}$ is attached to the class token to obtain $F^{\mathrm{T}}_{\mathrm{output}}$. Here, the class token inherits the statistical characteristics of other patch embeddings from the transformer input and contains the class-specific information. Features from $F^{\mathrm{C}}_{\mathrm{patch}}$ are rich in local features and bring guidance from CNN to transforms. After $F^{\mathrm{T}}_{\mathrm{output}}$ is input to the next transformer stage, the self-attention mechanism can explore global features based on the local features from the CNN to enhance the feature representation.

As shown in Fig. 3, only FMM-2 and FMM-4 have the output for the CNN feature extractor, i.e., $F^{\mathrm{C}}_{\mathrm{output}}$. Here, $F^{\mathrm{C}}_{\mathrm{output}}$ of FMM-2 and FMM-4 are input to IAM-1 and IAM-2, respectively. In the FMM, transformer input $F^{\mathrm{T}}_{\mathrm{input}}$ is resized to the image space to obtain $F^{\mathrm{T}}_{\mathrm{image}}$. $F^{\mathrm{C}}_{\mathrm{output}}$ is calculated as follows:

$$F^{\mathrm{C}}_{\mathrm{output}} = \sigma\left(F^{\mathrm{C}}_{\mathrm{input}}\right) \otimes F^{\mathrm{T}}_{\mathrm{image}} \tag{4}$$

where σ denotes the sigmoid function, and $\otimes$ represents the elementwise multiplication. The $\sigma(F^{\mathrm{C}}_{\mathrm{input}})$ works as a mask to provide suitable spatial support for $F^{\mathrm{T}}_{\mathrm{image}}$. $F^{\mathrm{T}}_{\mathrm{image}}$ is converted from transformer embeddings and rich in global features. $F^{\mathrm{C}}_{\mathrm{output}}$ can inherit global features from $F^{\mathrm{T}}_{\mathrm{input}}$, and these features are sent to the CNN feature extractor. With FMM, local features from the CNN feature extractor and global features from the transformer pathway are better retained during the conversion process.

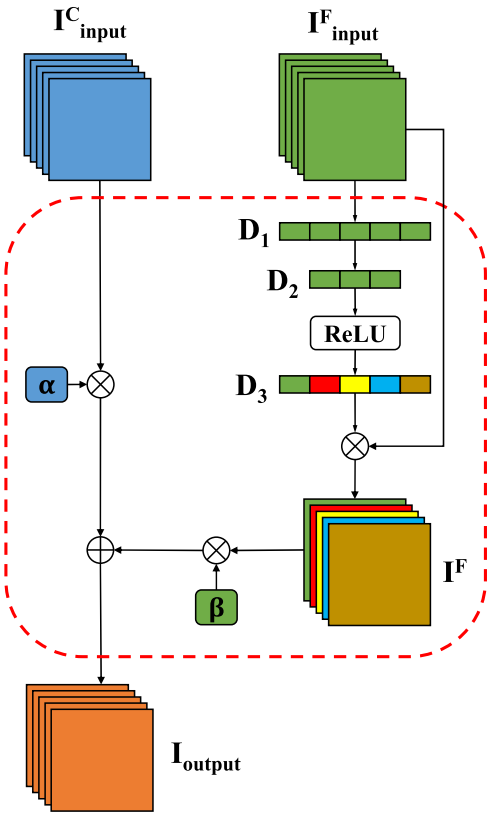


Fig. 6. Structure of IAM.

D. Intelligent Aggregation Module

When fed back to the CNN feature extractor, the transformer patch embeddings are aggregated with the CNN feature maps in the IAM. The IAM attempts to make the aggregation an intelligent process that can capture and select the most valuable information for prediction. As shown in Fig. 6, the IAM have two inputs, i.e., $I^{\mathrm{C}}_{\mathrm{input}}$ $(B \times C \times H \times W)$ and $I^{\mathrm{F}}_{\mathrm{input}}$ $(B \times C \times H \times W)$, where $I^{\mathrm{C}}_{\mathrm{input}}$ comes from the CNN stage and $I^{\mathrm{F}}_{\mathrm{input}}$ comes from the FMM. $I^{\mathrm{F}}_{\mathrm{input}}$ is the same as $F^{\mathrm{C}}_{\mathrm{output}}$ of the FMM and abundant in global features. Here, to select the most useful global features in $I^{\mathrm{F}}_{\mathrm{input}}$ and use these features for aggregation, we establish interdependencies between the

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channels of $I_{\text{input}}^{\text{F}}$. As shown in Fig. 6, these dependencies are reflected in descriptor D_3 of size $B \times C \times 1 \times 1$, where C is the number of channels of $I_{\text{input}}^{\text{F}}$. Note that there is only one pixel on each channel of D_3, and the value of each pixel represents the importance of the corresponding channel of $I_{\text{input}}^{\text{F}}$. Here, $I_{\text{input}}^{\text{F}}$ is multiplied D_3 to emphasize informative channels and suppress less informative channels [41]. To obtain D_3, global average pooling is first employed to aggregate the spatial information of $I_{\text{input}}^{\text{F}}$ and obtained descriptor D_1

$$D_1^k = \left[\sum_{i=1}^{H}\sum_{j=1}^{W}\left(I_{\text{input}}^{\text{F}}\right)^k(i,j)\right] \Big/ H/W \tag{5}$$

where the size of D_1 is the same as that of D_3. Here, D_1^k represents the kth element of D_1, and $(I_{\text{input}}^{\text{F}})^k(i,j)$ denotes the pixels on the kth channel of $I_{\text{input}}^{\text{F}}$. The global average pooling helps obtain a descriptor and strengthen the representation of global features. We then use a fully connected layer L_1 to compact D_1

$$D_2 = L_1(D_1) \tag{6}$$

where $D_2 \in B \times (C/r) \times 1 \times 1$, and r is the compact ratio (r is set to 16 in this article). After D_2 goes through a ReLU layer, another fully connected layer L_2 is employed to restore the channel number and make use of the information embedded in D_2

$$D_3 = \sigma(L_2(D_2)) \tag{7}$$

where σ is the sigmoid function. During training, the values of the elements in D_3 change dynamically in the compact-restore process, and D_3 can learn the interdependencies among the channels of $I_{\text{input}}^{\text{F}}$. Here, I^{F} is obtained by rescaling $I_{\text{input}}^{\text{F}}$ with $D3$ as follows:

$$I^{\text{F}} = I_{\text{input}}^{\text{F}} \cdot D3. \tag{8}$$

Attributed to descriptor D_3, channels with more beneficial information are further emphasized, and the feature representation of I^{F} is improved.

$I_{\text{input}}^{\text{C}}$ is obtained by the CNN feature extractor based on the pretrained model; thus, it contains class-specific information. However, the inherent structure of the CNN determines that $I_{\text{input}}^{\text{C}}$ contains more local features but lacks representations of the global features. Given that I^{F} is converted from transformer patch embeddings and is rich in global features, we integrate these global features to $I_{\text{input}}^{\text{C}}$ via aggregation. As a result, the representational power is enhanced, and the performance of the proposed TECN is improved. As shown in Fig. 6, the output of IAM is the weighted sum of $I_{\text{input}}^{\text{C}}$ and I^{F}

$$\left(I_{\text{output}}\right)^i = \alpha^i \cdot \left(I_{\text{input}}^{\text{C}}\right)^i + \beta^i \cdot \left(I^{\text{F}}\right)^i \tag{9}$$

where $(I_{\text{output}})^i$ denotes the feature map on the ith channel of I_{output}. $(I_{\text{input}}^{\text{C}})^i$ and $(I^{\text{F}})^i$ denote the ith feature channels of $I_{\text{input}}^{\text{C}}$ and I^{F}, respectively. Here, α and β are the sequences of real numbers, and the length of these sequences is the same as the number of channels in $I_{\text{input}}^{\text{C}}$. Note that the initial values of elements in α and β are set to 1. Like other parameters in the model, the elements in α and β are updated after each epoch during training. α and β determine the weights contributed by the features from $I_{\text{input}}^{\text{C}}$ and I^{F}, respectively. These trainable sequences make the aggregation an intelligent process. The statistical characteristics of $I_{\text{input}}^{\text{C}}$ and I^{F} differ from each other, e.g., there may be large differences in the mean and variance of the pixels in each channel. Thus, simple addition of $I_{\text{input}}^{\text{C}}$ and I^{F} can lead to a loss of local or global features. The intelligent aggregation can obtain the proportion of local features and global features on each feature map such that these two types of features are retained simultaneously. As a result, the global features from transformer can enhance the local features from the CNN, and the representation power of the proposed TECN is improved.

IV. Experiments

A. Dataset

In the RS field, researchers have developed several scene classification datasets, e.g., Aerial Image [42], UC Merced (UCM) land-use [43], and Northwestern Polytechnical University-REmote Sensing Image Scene Classification-45 classes (NWPU-RESISC45) [44]. These datasets primarily comprise land-use scenes, e.g., railways, stadiums, palaces, and harbors. Few RS classification datasets focus on smoke and smoke-like scenes. Smoke can be treated as an indicator of wildfire in RS images; thus, recognition of smoke in RS images is of great significance to early identification of wildfires, assessing trends in how fires spread, and the safety of human life. In addition, smoke differs visually from those land-use scenes; thus, deep learning models designed for smoke and land-use scenes will also differ. To fill the gap in this field, a previous work [1] developed the USTC_SmokeRS classification dataset for RS smoke recognition. This dataset contains six scenes, i.e., smoke, dust, haze, cloud, land, and seaside, where cloud, dust, and haze are smoke-like scenes, and land and seaside are the background where the other four scenes were captured. Note that dust and haze are similar to smoke. Clouds often appear in RS images and may be treated as smoke when wildfires occur. These high interclass similarities make it difficult to distinguish smoke from smoke-like scenes. The previous study [1] selected Moderate Resolution Imaging Spectroradiometer (MODIS) data as the source for the USTC_SmokeRS dataset. After performing geometric correction and radiometric calibration using the ENVI software, the 1, 4, and 3 bands of the MODIS data were selected as the red, green, and blue channels, respectively. The true-color composition RGB images generated from the MODIS 1, 4, and 3 bands were used to construct the images in the USTC_SmokeRS dataset. Note that the bandwidths of the MODIS 1, 4, and 3 bands are 0.620–0.67, 0.545–0.565, and 0.459–0.479 μm, respectively.

The USTC_SmokeRS dataset contains 6225 images, and several examples are shown in Fig. 7. As can be seen, smoke, dust, haze, and cloud are similar in color, texture, and spectral characteristics. The number of images in the dataset for each scene is given in Table II. The images in this dataset are captured over six continents (except for Antarctica), and the geographic distribution of the images is given in Table III. For

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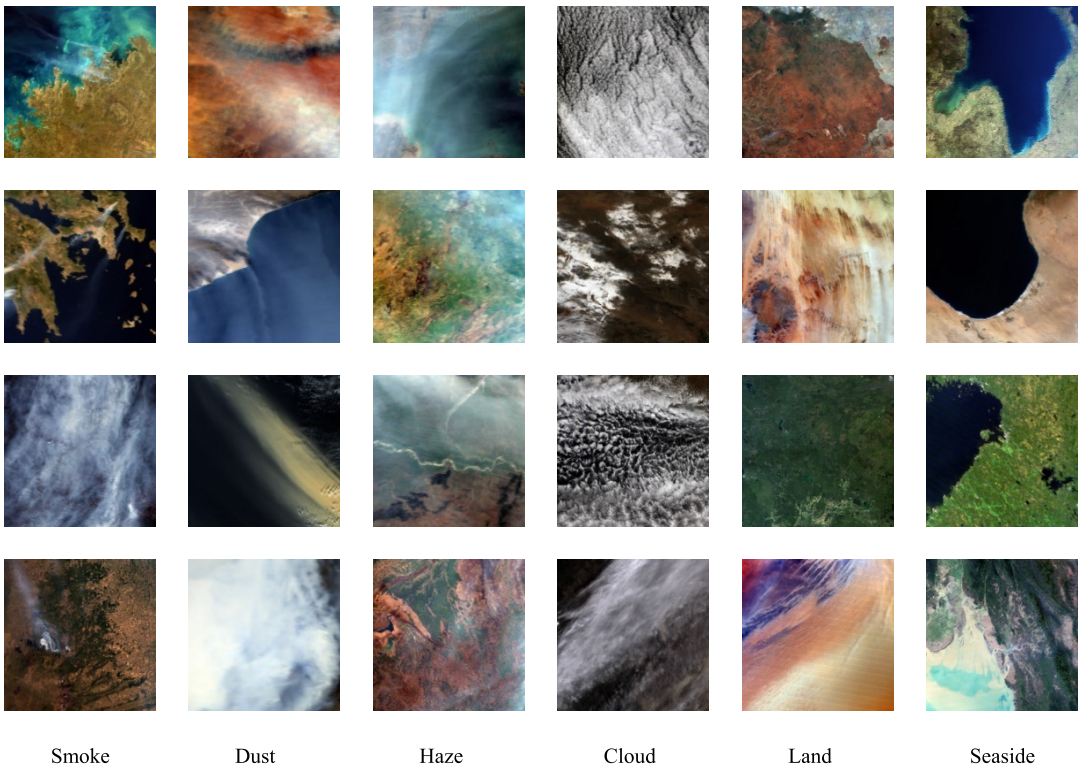


Fig. 7. Example images from USTC_SmokeRS dataset.

TABLE II
NUMBER OF IMAGES PER SCENE FOR USTC_SMOKERS

Scene	Smoke	Dust	Haze	Cloud	Land	Seaside	Total
Number	1016	1009	1002	1164	1027	1007	6225

TABLE III
GEOGRAPHIC DISTRIBUTION OF IMAGES IN USTC_SMOKERS DATASET

	Africa	Europe	Oceania	South America	North America	Asia
Smoke	0.7%	6.3%	10.5%	3.7%	59.7%	19.0%
Dust	35.6%	0.8%	1.3%	0.0%	0.0%	62.3%
Haze	0.0%	0.0%	0.0%	0.0%	0.0%	100.0%
Cloud	2.6%	5.1%	7.2%	1.1%	60.2%	23.8%
Land	6.6%	5.2%	21.8%	1.8%	51.9%	12.8%
Seaside	20.7%	11.2%	9.8%	2.0%	20.9%	35.5%

example, 59.7% of the smoke images are collected over North America due to the frequent occurrence of large wildfires in this region. The dust images are mostly obtained over Africa and Asia, and all haze images are acquired over Asia. Additional details about this dataset can be found in [1]. The 6225 images in the USTC_SmokeRS dataset are divided into a training set (3984%), a validation set (999; 16%), and a testing set (1242; 20%).

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TABLE IV
CLASSIFICATION ACCURACY OF DIFFERENT MODELS

Models	Accuracy (%)
VggNet-19-BN [3]	84.38
ResNet-34 [4]	85.53
ResNet-50 [4]	86.23
DenseNet-121 [5]	87.52
Pretrained-VggNet-19 [3]	91.47
Pretrained-ResNet-34 [4]	97.19
Pretrained-ResNet-50 [4]	97.74
Pretrained-DenseNet-121 [5]	97.98
SmokeNet [1]	92.75
HRNet-w64 [45]	94.28
BoCF [20]	94.91
RSSC-ETDL [25]	95.42
D-CNN [27]	96.03
KFBNet [29]	95.88
ViT-B [8]	89.77
Swin-B [32]	92.19
DeiT-B [33]	92.84
MobileViT-S [35]	94.52
Conformer [6]	92.43
TECN	**98.39**

B. Evaluation Metric

In this study, classification accuracy and the confusion matrix are selected as evaluation metrics, which are widely used to evaluate the performance of scene classification tasks. Classification accuracy is computed as the number of correctly predicted images divided by the total number of testing images. For the confusion matrix, each row represents true labels, and each column represents the predicted category for the given input.

C. Implementation Details

During the training process, the maximum number of epochs is set to 200. Here, random horizontal and vertical flipping are applied to the input images. All input images are normalized with a mean of (0.5, 0.5, 0.5) and a deviation of (0.5, 0.5, 0.5). To facilitate fair comparison, the above settings are the same as settings reported in [1]. Note that we use stochastic gradient descent (SGD) during training. In addition, cross-entropy loss is selected as the loss function. The initial learning rate is set to 0.01, which was reduced by a factor of 0.1 at the 70th and 140th epochs. PyTorch is used to implement the proposed TECN. For the CNN feature extractor, the pretrained weights of DenseNet-121 are taken from PyTorch's model library. All experiments are conducted on an Ubuntu 18.04 system with a single 3.2-GHz four-core i5-6500 CPU with 12-GB memory. A NVIDIA GTX 2070 GPU is used for acceleration. For fair comparison, each model is trained, validated, and tested on the same images under the same experimental conditions.

D. Performance

Extensive experiments are performed to demonstrate the effectiveness of the proposed TECN. The classification accuracies of different classification models are given in Table IV. As can be seen, the proposed TECN achieves the highest accuracy of 98.39%. VggNet-19 obtains the lowest accuracy because its structure is relatively simple. We find that ResNet-34 outperformed VggNet-19, which can be attributed to identity mapping. With an increased network depth, ResNet-50 further improves accuracy over ResNet-34. Benefited from dense connections, the vanishing gradient problem is further solved, and DenseNet-121 obtains a classification result of 87.52%. Note that the networks loaded with pretrained models in Table IV (Pretrained-VggNet-19, Pretrained ResNet-34, Pretrained ResNet-50, and Pretrained DenseNet-121) adopt parameters pretrained on ImageNet, which are then fine-tuned on the USTC_SmokeRS dataset. These pretrained parameters already contain scene information from ImageNet, and they significantly improve the performance of these networks. SmokeNet utilizes spatial and channelwise attention to capture features, and it is the baseline model of the USTC_SmokeRS dataset. High-resolution network (HRNet) maintains high-resolution representations through the entire network and obtains the accuracy of 94.28%. The bag of convolutional features (BoCF), RSSC-ETDL, D-CNN, and KFBNet models are all designed for the classification of RS land-use scenes, and they obtain remarkable accuracies. However, the natural gap between land-use and smoke-like scenes does not allow them to outperform the proposed TECN. Note that all of the above models are CNN-based models. Differing from CNN-based models, transformers use the self-attention mechanism to extract features. We find that the standard ViT-B model achieves 89.77% accuracy. Shifted windows and the hierarchical structure help Swin transformer-B improve accuracy based on ViT. The DeiT-B model introduces a new distillation procedure, and its accuracy is 92.84%. The MobileViT and conformer hybrid models integrate CNN and transformer models, and they achieve accuracies of 94.52% and 92.43%, respectively. The proposed TECN simultaneously retains local features from the CNN and global features from transformer. The FMM and IAM of the proposed TECN make the aggregation of different features smoother and more effective. Overall, the proposed TECN achieves the highest accuracy on the USTC_SmokeRS dataset.

The confusion matrix for the proposed TECN is given in Fig. 8, and the classification accuracies for the six scenes are listed in Table V. We find that all haze images were classified correctly. Although haze is white, it is also relatively thin and light compared to smoke, dust, and cloud, as shown in Fig. 7. This feature may be the primary reason why each haze image is predicted correctly. The accuracies obtained for cloud and seaside images are all greater than 99%. There are many cloud clusters in cloud images, and they are evenly distributed, which makes the texture of cloud images differs slightly from that of other scenes. For example, in seaside images, there are rare white pixels. In each seaside image, approximately half of each image shows sea, and the other half shows seashore. These characteristics make it easy to recognize seaside images. Note that two smoke images are misclassified as seaside images. The reason for this misclassification may be due to the fact that the highly reflective sea surface occasionally looks like smoke. We find that four smoke images are misclassified as haze images, which can be

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TABLE V
CLASSIFICATION ACCURACIES OF SIX SCENES BY THE PROPOSED TECN

Scene	Smoke	Dust	Haze	Cloud	Land	Seaside
Accuracy(%)	97.04	97.51	100	99.14	97.56	99.00



Fig. 8. Confusion matrix of TECN for USTC_SmokeRS.

attributed to the similarity between their colors and textures. In addition, two dust images are classified as smoke, and three dust images are classified incorrectly as haze. These results reflect the high interclass similarity among smoke, dust, and haze. The 97.04% classification accuracy of smoke scenes demonstrates that recognizing smoke from smoke-like scenes is difficult but necessary. The gradient-weighted class activation mapping (Grad-CAM) [46] is employed to evaluate the performance of the proposed TECN qualitatively. Grad-CAM can generate visualizations for input images and explain a model's decision. In the heat map generated by Grad-CAM, the depth of the red mask reflects the degree of importance of different regions in an image. The Grad-CAM visualization results for the proposed TECN are shown in Fig. 9. As can be seen, TECN accurately located the semantic information in the input images.

E. Performances on Other Datasets

Considering that the volume of the USTC_SmokeRS is relatively small, we conduct experiments on other four datasets to further prove the effectiveness of TECN. These four datasets are Expanded USTC_SmokeRS, Aerial Image dataset (AID) [42], UCM [43], and NWPU-RESISC45 [44]. The Expanded USTC_SmokeRS dataset is created based on the original USTC_SmokeRS by ourselves. The AID, UCM, and NWPU-RESISC45 are widely used datasets for RS scene classification. Different from USTC_SmokeRS, the AID, UCM, and NWPU-RESISC45 mainly consist of land-use scenes in RS images. Example images of AID, UCM, and NWPU-RESISC45 are shown in Fig. 10. Training details for these four datasets are the same as those performed on the original USTC_SmokeRS. The experimental results show that TECN can also achieve competitive accuracies on Expanded USTC_SmokeRS and these three RS land-use classification datasets.

1) Expanded USTC_SmokeRS: Cloud is very common in RS images and Li *et al.* [47] proposed an RS cloud detection dataset. We use this dataset to enlarge the original USTC_SmokeRS and name the new dataset Expanded USTC_SmokeRS. This RS cloud detection dataset [47] is created using the GaoFen-1 Level-1A and ZiYuan-3 imagery. Its testing set contains 25 large GaoFen-1 and ZiYuan-3 images with pixel-level annotation labels. Its validation set consists of five GaoFen-1 images with pixel-level annotation labels. Resolutions of these images range from 4000×5000 to 9000×9000 pixels, and annotation labels determine whether each pixel contains clouds or not. We crop these large images to 256×256 pixels image patches to fit the image size of USTC_SmokeRS dataset. If annotation labels indicate there is no cloud pixel in one cropped image patch, this patch will be dropped. Considering the relative balance between different scenes, we select 3000 cropped patches with cloud pixels and add these patches to the category "Cloud" of the original USTC_SmokeRS dataset. Some example patches are shown in Fig. 11. Then, we obtain the Expanded USTC_SmokeRS dataset. Images of categories "Smoke," "Dust," "Haze," "Land," and "Seaside" in the Expanded USTC_SmokeRS are exactly the same as those in the original USTC_SmokeRS. Images belonging to the category "Cloud" in the new dataset contain two parts: one part comes from the "Cloud" category in the original USTC_SmokeRS, and the other part is the 3000 cropped images. The number of images per category for Expanded USTC_SmokeRS is listed in Table VI. Proportions of the training set, validation set, and testing set are 64%, 16%, and 20%, respectively, which are the same as those of USTC_SmokeRS. We compare TECN with other algorithms on the Expanded USTC_SmokeRS, and the results are listed in Table VII. As can be seen, pretrained DenseNet-121 achieves the accuracy 98.05%. TECN surpasses DenseNet-121 by 0.92% and achieves the highest accuracy 98.97%. This result can further prove the effectiveness of the proposed TECN. Confusion matrix of TECN for Expanded USTC_SmokeRS is shown in Fig. 12.

2) AID: AID is a large-scale dataset for RS scene classification and has 10 000 images in total. Image size is $3 \times 600 \times 600$ and the spatial resolution varies from 0.5 to 8 m. There are 30 scenes in AID: airport, bare land, baseball field, beach, bridge, center, church, commercial, dense residential, desert, farmland, forest, industrial, meadow, medium residential, mountain, park, parking, playground, pond, port, railway station, resort, river, school, sparse residential, square, stadium, storage tanks, and viaduct. Each scene contains 220–420 images. Following previous works on AID, we

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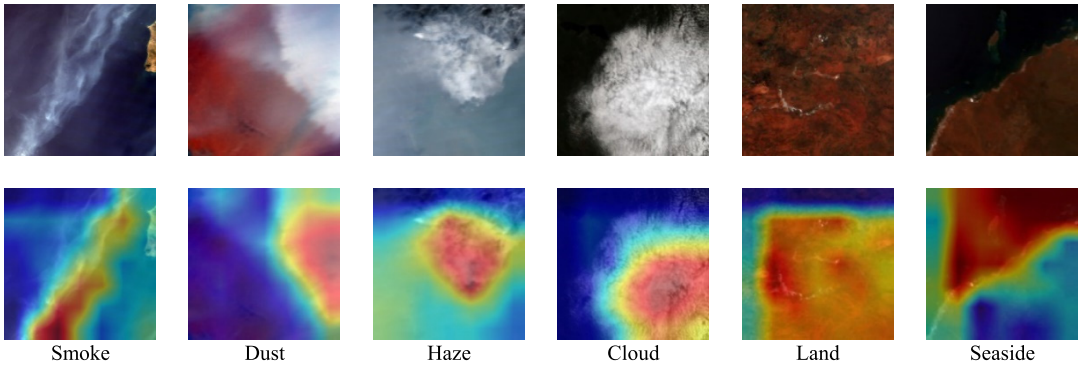


Fig. 9. Visualization results for TECN.



Fig. 10. Examples images of AID, UCM, and NWPU-RESISC45.

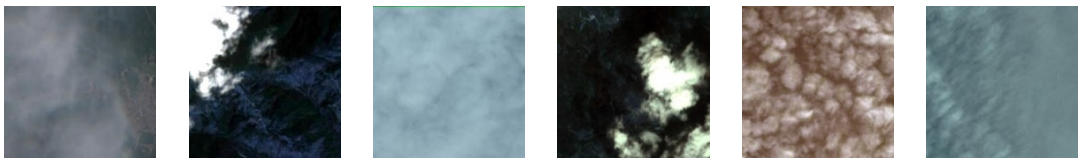


Fig. 11. Examples of cropped cloud patches.

TABLE VI
NUMBER OF IMAGES PER SCENE FOR EXPANDED USTC_SMOKERS

Scene	Smoke	Dust	Haze	Cloud	Land	Seaside	Total
Number	1016	1009	1002	4164	1027	1007	9225

randomly select 20% or 50% data for training and the rest are used for testing. Classification accuracies on AID are listed in Table VIII, and TECN shows remarkable performances. When the training rate is 20%, TECN achieves 95.45% accuracy, which is very close to the state-of-the-art (SOTA) performance 95.50%. When training rate is 50%, TECN obtains 97.40% accuracy, which surpasses pretrained-DenseNet121 by 1.6%, and is equal to the highest accuracy 97.40%. Confusion matrix of TECN for AID with 50% training rate is shown in Fig. 13.

3) UCM: UCM land-use dataset is the first ground-truth dataset set up using available RS images. UCM contains 2100 images and image size is $3 \times 256 \times 256$. There are 21 scenes and each scene has 100 images. The 21 scenes are agricultural, airplane, baseball diamond, beach, buildings,

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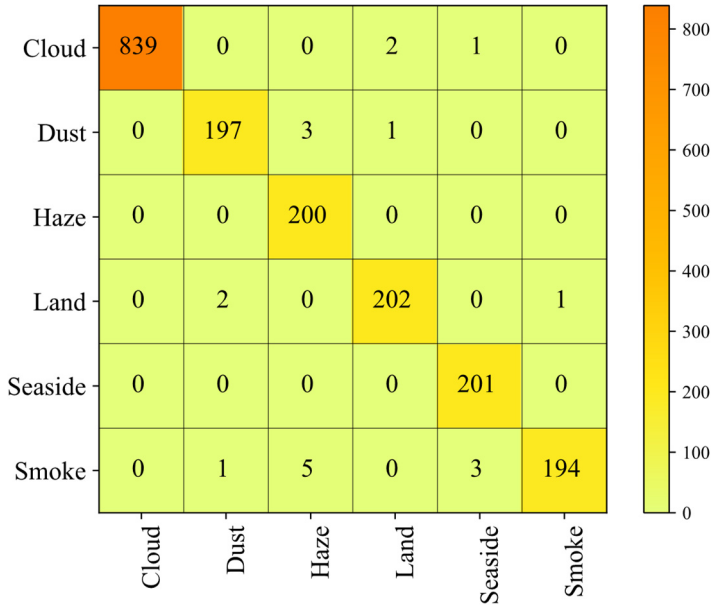


Fig. 12. Confusion matrix of TECN for expanded USTC_SmokeRS.

TABLE VII
CLASSIFICATION ACCURACIES ON EXPANDED USTC_SMOKERS

Models	Accuracy (%)
BoCF [20]	96.74
RSSC-ETDL [25]	97.89
SmokeNet [1]	95.41
Pretrained DenseNet121[5]	98.05
Swin-B [32]	93.95
MobileViT-S [35]	96.92
KFBNet [29]	98.08
TECN	**98.97**

TABLE VIII
CLASSIFICATION ACCURACIES ON AID

Models	Accuracy (%)	
	T.R.=20%	T.R.=50%
BoCF [20]	90.50	95.93
RSSC-ETDL [25]	88.23	91.26
SmokeNet [1]	83.71	85.34
Pretrained DenseNet121[5]	93.93	95.80
Swin-B [32]	75.23	85.72
MobileViT-S [35]	83.33	92.66
KFBNet [29]	**95.50**	**97.40**
TECN	95.45	**97.40**

chaparral, dense residential, forest, freeway, golf course, harbor, intersection, medium residential, mobile home park, overpass, parking lot, river, runway, sparse residential, storage tanks, and tennis court. The spatial resolution is 0.3 m. Following existing methods, we randomly select 80% data in each category for training and the rest for testing. The experimental results on UCM are listed in Table IX. Because there

TABLE IX
CLASSIFICATION ACCURACIES ON UCM

Models	Accuracy (%)
BoCF [20]	96.90
RSSC-ETDL [25]	98.10
SmokeNet [1]	94.38
Pretrained DenseNet121[5]	97.86
Swin-B [32]	87.86
MobileViT-S [35]	95.24
KFBNet [29]	**99.88**
TECN	99.52

are only 2100 images and training rate is 80%, accuracies of different models on this dataset are relatively high. Pretrained-DenseNet121 gets a 97.86% accuracy. The SOTA performance 99.88% is achieved by KFBNet. TECN obtains an acceptable accuracy of 99.52%, while testing set contains 420 images. It can be seen from confusion matrix in Fig. 14, and only two images are misclassified by TECN.

4) NWPU-RESISC45: As far as we know, NWPU-RESISC45 is the largest RS scene classification dataset. It contains 31 500 images: there are 45 scenes and each scene has 700 images. Image size is $3 \times 256 \times 256$ and the spatial resolution is 0.2 m per pixel. The 45 scenes are airplane, airport, baseball diamond, basketball court, beach, bridge, chaparral, church, circular farmland, cloud, commercial area, dense residential, desert, forest, freeway, golf course, ground track field, harbor, industrial area, intersection, island, lake, meadow, medium residential, mobile home park, mountain, overpass, palace, parking lot, railway, railway station, rectangular farmland, river, roundabout, runway, sea ice, ship, snow berg, sparse residential, stadium, storage tank, tennis court, terrace, thermal power station, and wetland. It is a challenging dataset because several scenes are really similar

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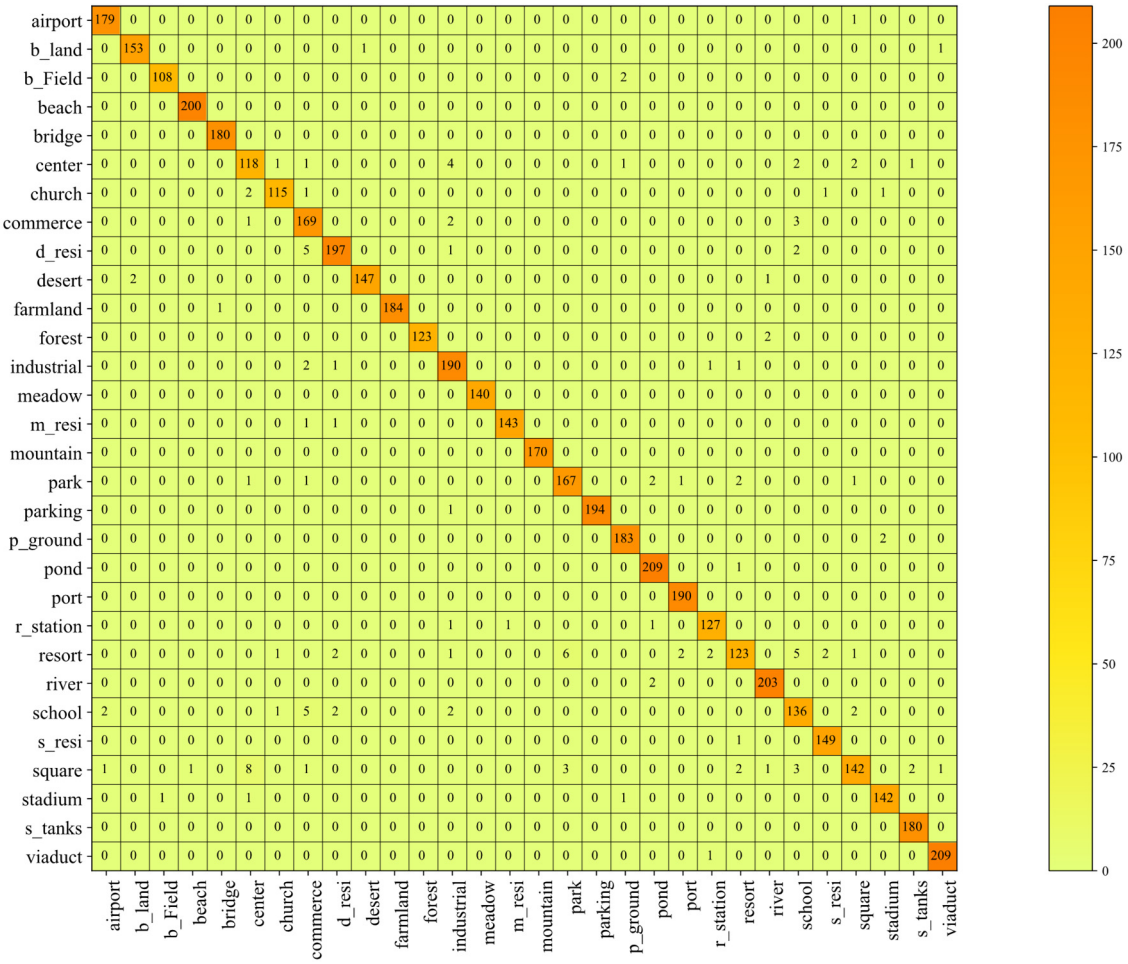


Fig. 13. Confusion matrix of TECN for AID with 50% training rate.

TABLE X
CLASSIFICATION ACCURACIES ON NWPU-RESISC45

Models	Accuracy (%)	
	T.R.=10%	T.R.=20%
BoCF [20]	90.96	92.14
RSSC-ETDL [25]	87.15	89.87
SmokeNet [1]	79.03	81.67
Pretrained DenseNet121[5]	90.20	92.97
Swin-B [32]	69.92	76.71
MobileViT-S [35]	77.35	85.15
KFBNet [29]	**93.08**	**95.11**
TECN	93.05	95.08

with each other. Following previous works, we use 10% or 20% data for training and the rest for testing. Results on this dataset are shown in Table X. When the training rate is 10%, the highest accuracy 93.08% is obtained by KFBNet. TECN achieves the accuracy of 93.05%, 0.03% lower than the highest one. Testing set has 28 350 images under 10% training rate, and the number of correctly recognized images by TECN is only 8 less than that of KFBNet. When training rate is 20%, the highest result is also obtained by KFBNet as 95.11%. TECN achieves a 95.08% accuracy. Considering there are 25 200 images in testing set under 20% training rate, the number of correctly classified images by TECN is 7 less than that of KFBNet. Confusion matrix of TECN for NWPU-RESISC45 with 20% training rate is shown in Fig. 15.

Performances on these datasets prove the robustness and effectiveness of TECN again. On Expanded USTC_SmokeRS, TECN obtains the highest accuracy. TECN also achieves competitive accuracies on RS land-use datasets AID, UCM, and NWPU-RESISC45, while TECN is specially designed for smoke-like scenes.

F. Implementation Speed

We also compare the number of parameters and implementation speed of the proposed TECN to those of other models. During the inference, the size of the input image is set to $4 \times 3 \times 256 \times 256$, and we compute the average time required for each image. The results are given in Table XI. We find that the proposed TECN only consumes 28.4 ms for each image and realizes the real-time performance.

V. ABLATION STUDY

We also conduct ablation studies on USTC_SmokeRS to verify the rationality of the proposed TECN.

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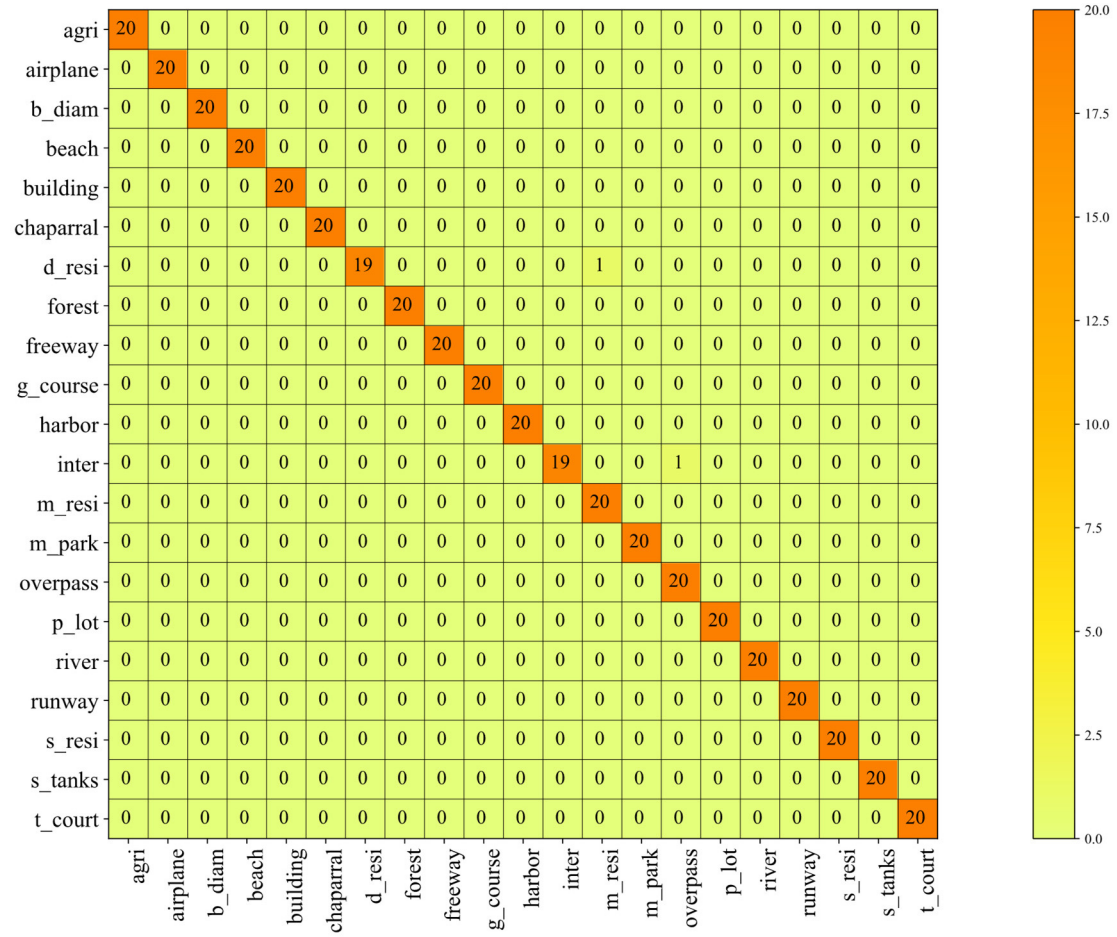


Fig. 14. Confusion matrix of TECN for UCM.

TABLE XI
IMPLEMENTATION COSTS OF DIFFERENT MODELS

Models	#Params (M)	GFLOPs (G)	Speed/image (ms)
VggNet-19 [3]	143.7	19.6	4.9
ResNet-34 [4]	21.8	4.8	**2.5**
ResNet-50 [4]	25.6	5.4	4.4
DenseNet-121 [5]	8.0	11.5	10.9
SmokeNet [1]	53.5	14.7	15.3
HRNet-w64 [45]	128.1	29.0	25.4
BoCF [20]	36.9	8.9	8.4
RSSC-ETDL [25]	209.3	47.3	56.7
D-CNN [27]	74.1	16.6	18.2
KFBNet [29]	216.3	41.1	36.7
ViT-B [8]	86.0	55.4	11.6
Swin-B [32]	88.0	15.4	3.6
DeiT-B [33]	86.1	17.5	3.4
MobileViT-S [35]	**5.6**	**1.7**	3.6
Conformer [6]	83.3	23.3	26.1
TECN	146.9	24.9	28.4

A. Estimation of FMM and IAM

The purpose of FMM is to retain as many useful local features as possible in the process of converting CNN feature maps to transformer patch embeddings. In addition, the IAM is designed to aggregate local features from the CNN and global features from the transformer pathway. To demonstrate the effectiveness of the FMM and IAM, we compare three models, i.e., TECN without FMM, TECN without IAM, and the

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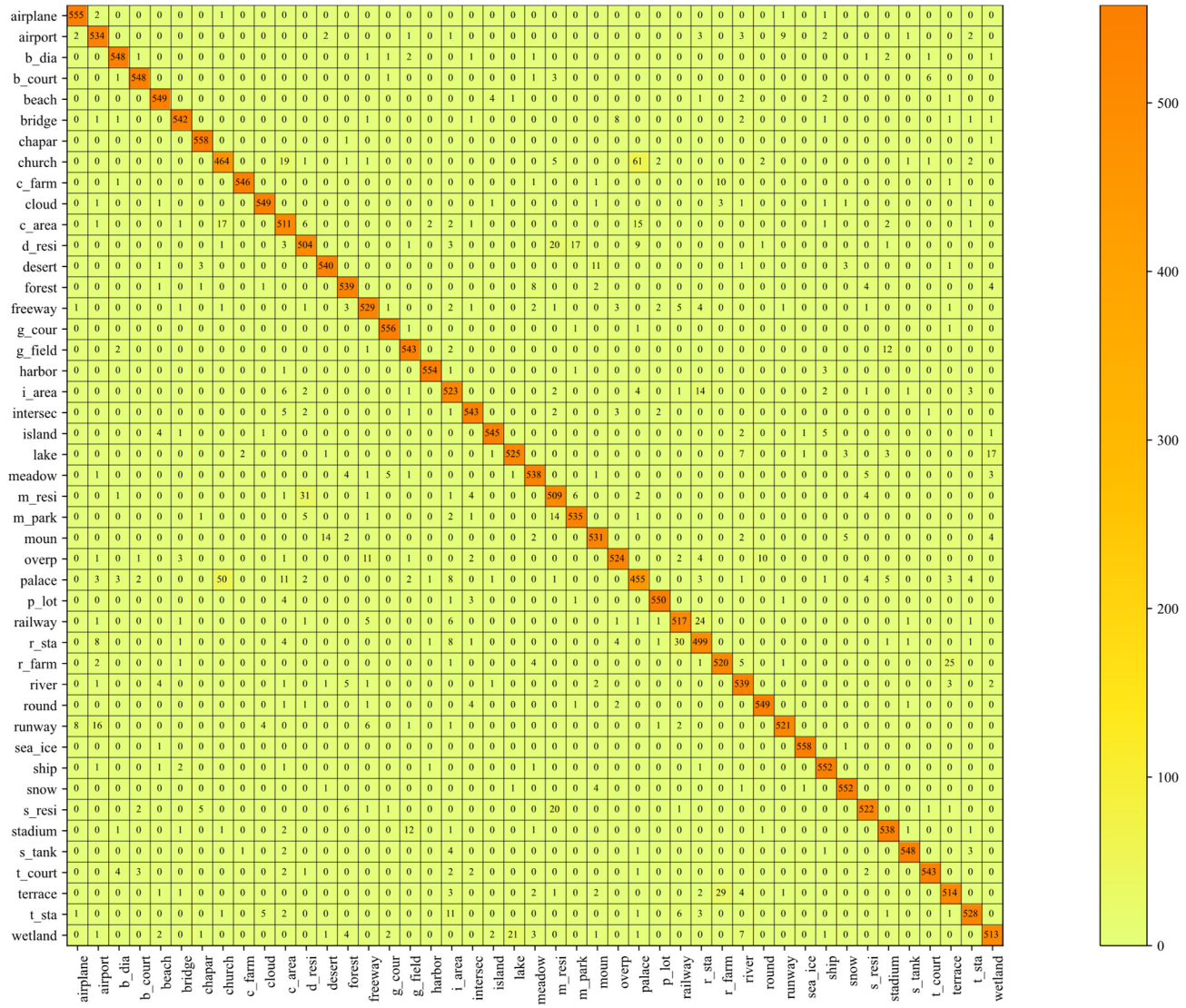


Fig. 15. Confusion matrix of TECN for NWPU-RESISC45 with 20% training rate.

TABLE XII
PERFORMANCE WITH AND WITHOUT FMM OR IAM

Models	Accuracy (%)
TECN w/o FMM	97.50
TECN w/o IAM	98.07
TECN	**98.39**

TABLE XIII
COMPARISONS OF DIFFERENT CNN FEATURE EXTRACTORS

Models	Accuracy (%)
Pretrained ResNet-34	97.19
TECN (ResNet-34)	97.76
Pretrained ResNet-50	97.74
TECN (ResNet-50)	98.12
Pretrained DenseNet-121	97.98
TECN	**98.39**

proposed TECN. The results given in Table XII demonstrate that both the FMM and IAM contribute to the proposed model's performance improvement.

B. Different CNN Feature Extractor Structures

In the scene classification task, some CNNs, e.g., ResNet, can achieve notable performance individually. In this study, the proposed TECN uses DenseNet-121 pretrained on ImageNet as the feature extractor. Here, we prove that the proposed TECN can function effectively with other feature extractors. In Table XIII, the TECN (ResNet-34) represents TECN uses pretrained ResNet-34 instead of DenseNet-121 as CNN feature extractor, and TECN (ResNet-50) is defined in the same way. As shown in Table XIII, the pretrained DenseNet-121 helps the proposed TECN achieve the highest accuracy, i.e., 98.39%; thus, we select the pretrained DenseNet-121 as the feature extractor for TECN. Although TECN (ResNet-34)

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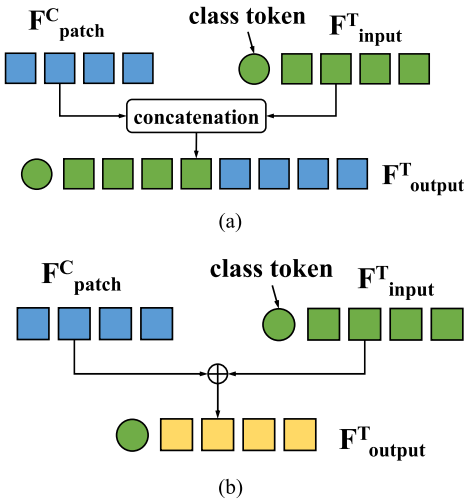


Fig. 16. Fusion modes in FMM: (a) concatenation and (b) addition.

TABLE XIV
COMPARISONS OF DIFFERENT TRANSFORMER PATHWAYS

Models	Accuracy (%)
TECN (ViT)	94.77
TECN (Swin)	95.49
TECN (DeiT)	95.13
TECN	**98.39**

and TECN (ResNet-50) do not outperform the proposed TECN, they do outperform the other original CNN models. TECN (ResNet-34) improves the accuracy of the Pretrained ResNet-34 by 0.57%, and TECN (ResNet-50) improves the accuracy of the Pretrained ResNet-50 by 0.38%. These results indicate the robustness of the TECN.

C. Different Structures of Transformer Pathway

Transformer has powerful feature representation in vision tasks, and several effective ViT backbones have been proposed. We implement and evaluate different structures for the transformer pathway, and results are given in Table XIV. Here, TECN (Swin) means that the original transformer pathway was replaced with the Swin transformer, and TECN (ViT) and TECN (DeiT) mean that the original transformer pathways were replaced by the ViT and DeiT transformers, respectively. The fact that the proposed TECN outperforms other models and verifies the effectiveness of the transformer pathway design in TECN.

D. Fusion Mode in FMM

As shown in Fig. 5, $F^{\mathrm{C}}_{\mathrm{patch}}$ and $F^{\mathrm{T}}_{\mathrm{input}}$ are fused in FMM. In this fusion process, all embeddings of $F^{\mathrm{T}}_{\mathrm{input}}$ are abandoned (except the class token). $F^{\mathrm{C}}_{\mathrm{patch}}$ is attached to the class token to obtain the $F^{\mathrm{T}}_{\mathrm{output}}$. In addition to the fusion method employed in the FMM, we also investigate two additional fusion modes, i.e., concatenation and addition, which are illustrated in Fig. 16. In the concatenation mode, $F^{\mathrm{C}}_{\mathrm{patch}}$ is concatenated with $F^{\mathrm{T}}_{\mathrm{input}}$ to obtain the $F^{\mathrm{T}}_{\mathrm{output}}$. For the addition mode, the patch embeddings from $F^{\mathrm{C}}_{\mathrm{patch}}$ and $F^{\mathrm{T}}_{\mathrm{input}}$ are added up at the corresponding position while retaining the class token. The classification results obtained with these fusion modes are shown in Table XV. We find that the concatenation method provides no benefit and increases computational costs. Overall, the fusion mode employed in the proposed TECN achieves the highest accuracy.

TABLE XV
DIFFERENT FUSION MODES IN THE FMM

Modes	Accuracy (%)
Concatenation	97.91
Addition	98.07
TECN	**98.39**

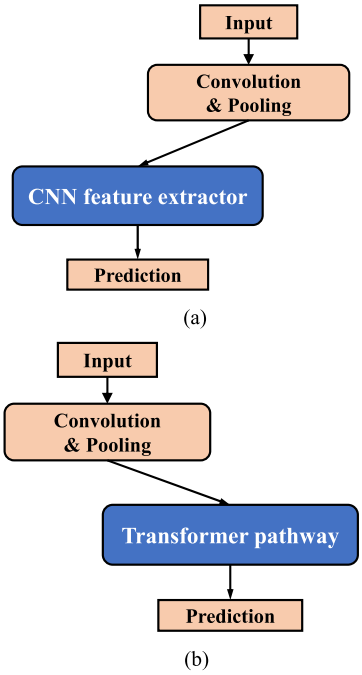


Fig. 17. (a) Only CNN feature extractor or (b) only transformer pathway.

E. Using Only CNN Feature Extractor or Transformer Pathway

The CNN feature extractor or transformer pathway of the proposed TECN can be used as an independent model to be trained on the dataset. The structures of these independent models are illustrated in Fig. 17. We investigate the performances of these independent models when working individually, and the obtained classification accuracies are shown in Table XVI. The CNN pathway alone achieves an accuracy of 97.39%, which is less than the reported accuracy of original pretrained DenseNet-121 by 0.59%. This can be attributed to the differences in structures of CNN pathway and DenseNet-121. As illustrated in Fig. 3, the CNN pathway is shown in the blue dashed line and it contains two IAMs:

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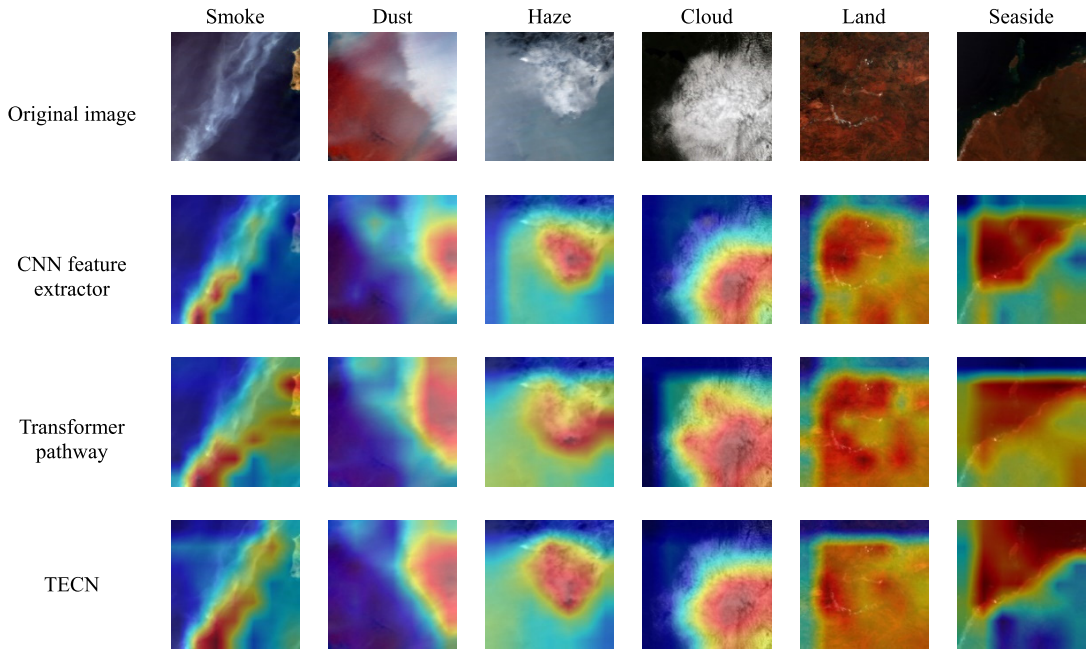


Fig. 18. Heat maps generated by different models.

TABLE XVI
CNN FEATURE EXTRACTOR OR TRANSFORMER PATHWAY WORKS INDIVIDUALLY

Models	Accuracy (%)
Only CNN	97.39
Only Transformer	96.78
TECN	**98.39**

TABLE XVII
ACCURACIES OF DIFFERENT VARIANTS

Variants	Accuracy (%)
Variant-1	97.79
Variant-2	96.98
Variant-3	98.06
TECN	**98.39**

IAM-1 and IAM-2. In contrast, there is no IAM in the structure of DenseNet. In CNN pathway, weights for CNN stage1, CNN stage2, CNN stage3, and CNN stage4 are taken from the pretrained weight parameters of DenseNet-121 on ImageNet. However, IAM-1 and IAM-2 are not initialized with pretrained parameters. The proposed TECN combines the advantages of both the CNN and transformer, and it outperforms both of the independent models. We use Grad-CAM [46] to obtain heat maps generated by the independent CNN feature extractor, the independent transformer pathway, and the proposed TECN (Fig. 18). For the same image, the red area in the heat map of transformer pathway is larger than that of the CNN feature extractor. This phenomenon is consistent with the fact that CNN is effective at capturing local features and the transformers focus more on global features. The area of red regions in the heat map of TECN is between the condition of CNN feature extractor and transformer pathway.

F. Some Variants of TECN

We also investigate some other variants of TECN. These variants are illustrated in Fig. 19, where structural differences between the variants and TECN are highlighted using red arrows. Results in Table XVII indicate that these variants achieved lower accuracies than the proposed TECN. In variant-1, there are shortcuts from the FMM to all four stages in the CNN feature extractors. It is known that transformers lack some of the inductive biases inherent to CNN and, therefore, cannot generalize well when training data are not sufficient [8]. The condition is that the volume of USTC_SmokeRS which consists of 6225 images is relatively small, so we should introduce more features from CNN pathway in the model. Results in Table XVI also indicate that CNN pathway alone surpasses transformer pathway alone by 0.61%. In variant-1, too many features from transformer pathway are introduced to the final output. Thus, variant-1 gets an accuracy of 97.79%, which is lower than that of TECN. For variant-2, its structure is too simple, and CNN and transformer pathway are only connected after stage4, so local features and global features cannot be fully fused. Thus, variant-2 achieves an accuracy of 96.98%, which is the lowest among these variants. For variant-3, there are shortcuts from CNN to transformer pathway. In these shortcuts, CNN feature maps are converted to transformer patches directly. On the other hand, in the designed FMM, the conversion from feature maps to patches includes several steps: global pooling, average pooling, and reshaping. Compared to conversions in FMM, shortcuts in variant-3 are too rigid and rough. As a result, these shortcuts degrade the model's performance and variant-3 obtains the

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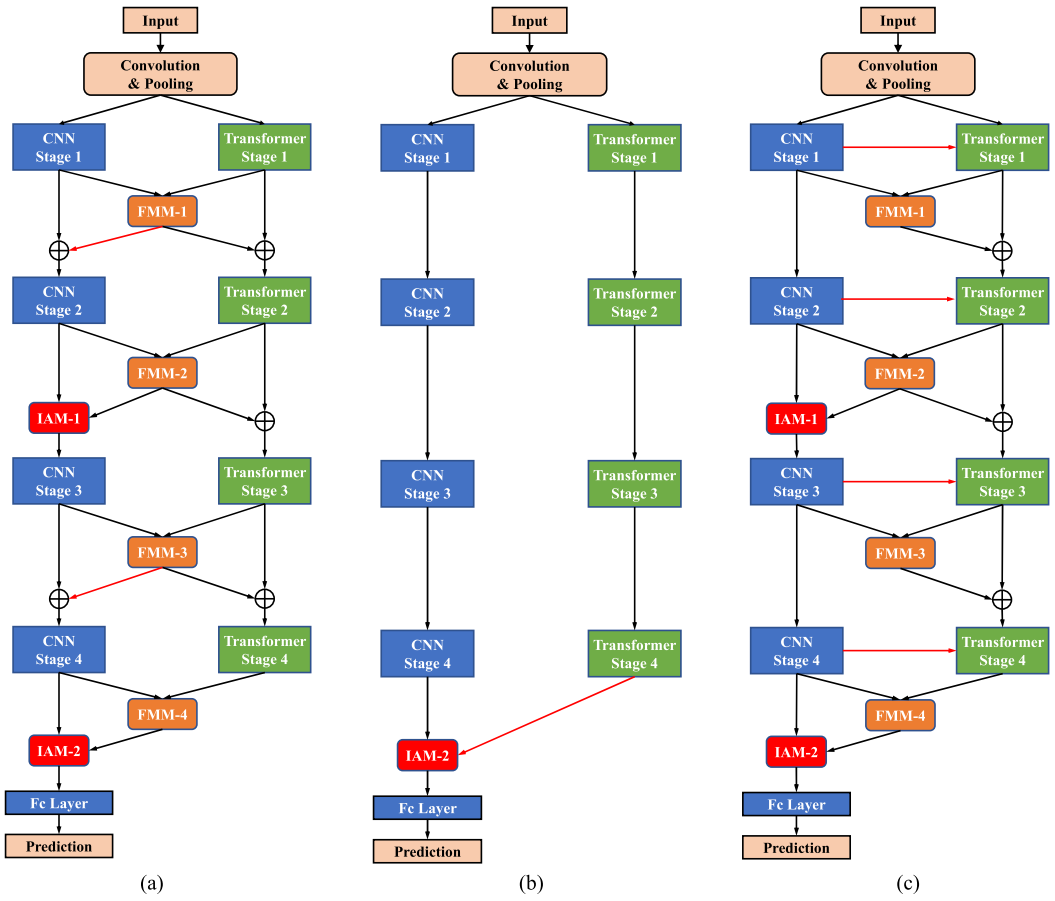


Fig. 19. Some variants of TECN: (a) variant-1; (b) variant-2; and (c) variant-3.

accuracy of 98.06%, slightly lower than the accuracy achieved by TECN (98.39%).

VI. Conclusion

In this article, we propose the TECN model for the classification of RS smoke-like scenes. CNNs excel at capturing local features, and transformers are effective at capturing global features. The proposed TECN bridges the CNN and transformer, and exploits the advantages of both. The proposed TECN includes two modules, i.e., the FMM and IAM, to enhance the feature representation abilities. The FMM helps retain local features when converting feature maps to patch embeddings, and the IAM makes the aggregation of feature maps and patch embeddings more effective.

In an extensive set of experiments, we compare the proposed TECN with other classification models on the USTC_SmokeRS dataset. The experimental results demonstrate that the proposed TECN achieves a competitive accuracy of 98.39%. On other larger RS dataset, TECN also shows strong robustness and effectiveness.

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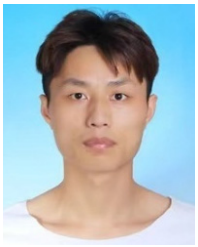
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Shikun Chen received the B.S. and M.S. degrees from the School of Mechanical Engineering, Shanghai Jiao Tong University, Shanghai, China, in 2015 and 2019, respectively. He is currently pursuing the Ph.D. degree with the School of Automation, Southeast University, Nanjing, China, and the Key Laboratory of Measurement and Control of Complex Systems of Engineering, Ministry of Education, Nanjing.

His research interests include computer vision, remote sensing image processing, and deep learning.



Weixuan Li received the B.S. degree from Jinling College, Nanjing University, Nanjing, China, in 2017.

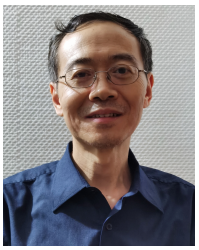
He currently works as an Image Processing Algorithm Engineer with Nanjing Enbo Technology Company Ltd., Nanjing. His research interests include fire detection and deep learning.



Yichao Cao received the M.S. degree from the School of Automation, Southeast University, Nanjing, Jiangsu, China, in 2019, where he is currently pursuing the Ph.D. degree with the Department of Automation and the Key Laboratory of Measurement and Control of Complex Systems of Engineering, Ministry of Education, Nanjing.

His research interests include image processing, deep learning, and computer vision.

Mr. Cao is a reviewer for international journals, such as *Neural Networks* and IEEE TRANSACTIONS ON COGNITIVE AND DEVELOPMENTAL SYSTEMS.



Xiaobo Lu is currently a Professor with the School of Automation, Southeast University, Nanjing, China, and the Key Laboratory of Measurement and Control of Complex Systems of Engineering, Ministry of Education, Nanjing. He has coauthored the book *An Introduction to the Intelligent Transportation Systems* (Beijing: China Communications Press, 2008). His research interests include image processing, signal processing, pattern recognition, and computer vision.

Prof. Lu received many research awards, such as the First Prize in Natural Science Award of the Ministry of Education of China and the prize in Science and Technology Award of Jiangsu province.

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